Attention Is All You Need

The Transformer Architecture

Vaswani et al. (2017) | *NeurIPS 2017* | *arXiv:1706.03762*

Overview

The Transformer architecture, introduced in 'Attention Is All You Need' (Vaswani et al., 2017), revolutionized natural language processing by replacing recurrent and convolutional layers entirely with attention mechanisms. This document summarizes the core concepts, architecture, and impact.

1. The Attention Mechanism

Self-attention allows a model to weigh the importance of different positions in the input sequence relative to each other. The core formula is:

```
Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) * V
```

where Q (queries), K (keys), and V (values) are linear projections of the input. The scaling factor 1/sqrt(d_k) prevents the dot products from growing too large in magnitude, which would push gradients into regions of the softmax where they become extremely small.

2. Multi-Head Attention

Rather than performing a single attention function, the Transformer uses multi-head attention, which runs h parallel attention operations with different learned projections. Each head can attend to information from different representation subspaces at different positions.

```
MultiHead(Q,K,V) = Concat(head_1, ..., head_h) * W^O
```

```
where head_i = Attention(Q*W_i^Q, K*W_i^K, V*W_i^V)
```

3. Encoder-Decoder Architecture

The Transformer consists of an encoder stack and a decoder stack, each composed of N=6 identical layers. Each encoder layer has two sub-layers: a multi-head self-attention mechanism, and a position-wise fully connected feed-forward network. Each decoder layer adds a third sub-layer performing multi-head attention over the encoder output.

Table 1: Transformer Model Configurations

Component	Dimensions (Base)	Dimensions (Large)

Model dimension (d_model)	512	1024
Feed-forward dimension	2048	4096
Number of heads (h)	8	16
Number of layers (N)	6	6
Key/Value dimension (d_k)	64	64
Parameters	65M	213M

4. Positional Encoding

Since the model contains no recurrence and no convolution, positional encodings are added to the input embeddings to inject information about the relative or absolute position of the tokens. The original paper uses sine and cosine functions of different frequencies:

```
PE(pos, 2i) = sin(pos / 10000^(2i/d_model))
```

```
PE(pos, 2i+1) = cos(pos / 10000^(2i/d_model))
```

5. Training & Results

The base model was trained on the WMT 2014 English-German dataset (~4.5M sentence pairs) and WMT 2014 English-French dataset (~36M sentence pairs). Training used 8 NVIDIA P100 GPUs. The Transformer achieved 28.4 BLEU on EN-DE (surpassing ensemble models) and 41.0 BLEU on EN-FR, establishing new state-of-the-art at a fraction of the training cost of previous models.

6. Impact and Legacy

- Formed the basis for BERT, GPT, T5, and virtually all modern LLMs
- Parallelizable training — no sequential bottleneck of RNNs
- Scales effectively: larger models consistently improve performance
- Extended beyond NLP to vision (ViT), audio (Whisper), and multimodal models
- Attention visualizations provide interpretability into model behavior

Reference

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I. (2017). Attention Is All You Need. NeurIPS 2017. arXiv:1706.03762